

Deterministic Coupling for Distributed Simulations

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Synchronization Bottleneck

- **Large-Scale PDES:** Parallel Discrete Event Simulations face scalability limits at partition boundaries.
- **Conservative Synchronization:** Prevents errors by blocking execution until all network messages arrive.
- **Optimistic Synchronization:** Speculatively executes while saving snapshots and rolls back if wrong

Rule-Based Coupling

- **Memoryless:** No historical snapshots; nodes assume boundary integrity during computation.
- **Asynchronous:** Nodes exchange boundary states after the compute phase.
- **Deterministic Priority Hierarchy:** A strict, globally shared tie-breaker resolves conflicts locally.

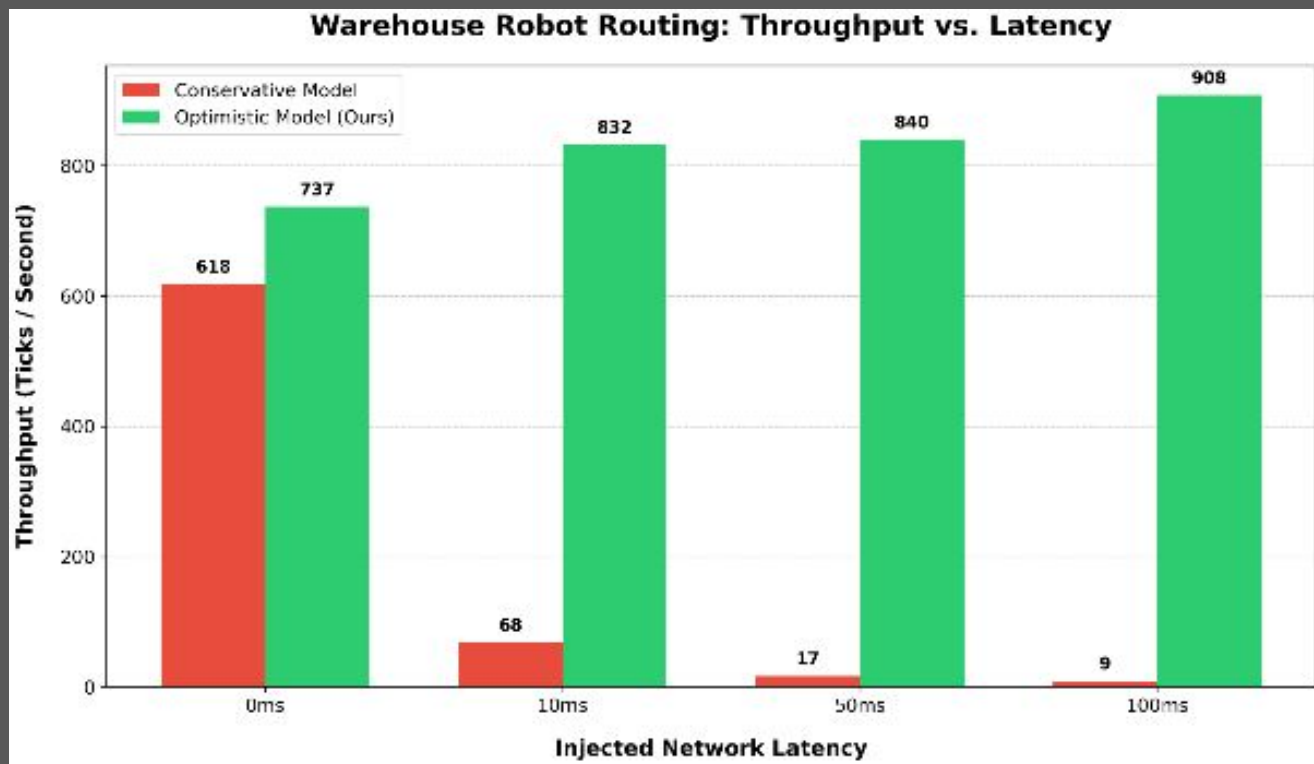
Implementation

- **Architecture:** Python master-worker model orchestrated via gRPC.
- **Thread Decoupling:** The Worker nodes separate the Simulation Logic (Main Thread) from the Network Hash Exchange (Background Thread)
- **Latency Injection:** Master dictates a network delay (e.g., 50ms). Worker's background thread deliberately sleeps before transmitting, simulating geographic network lag without dropping packets.

Simulations

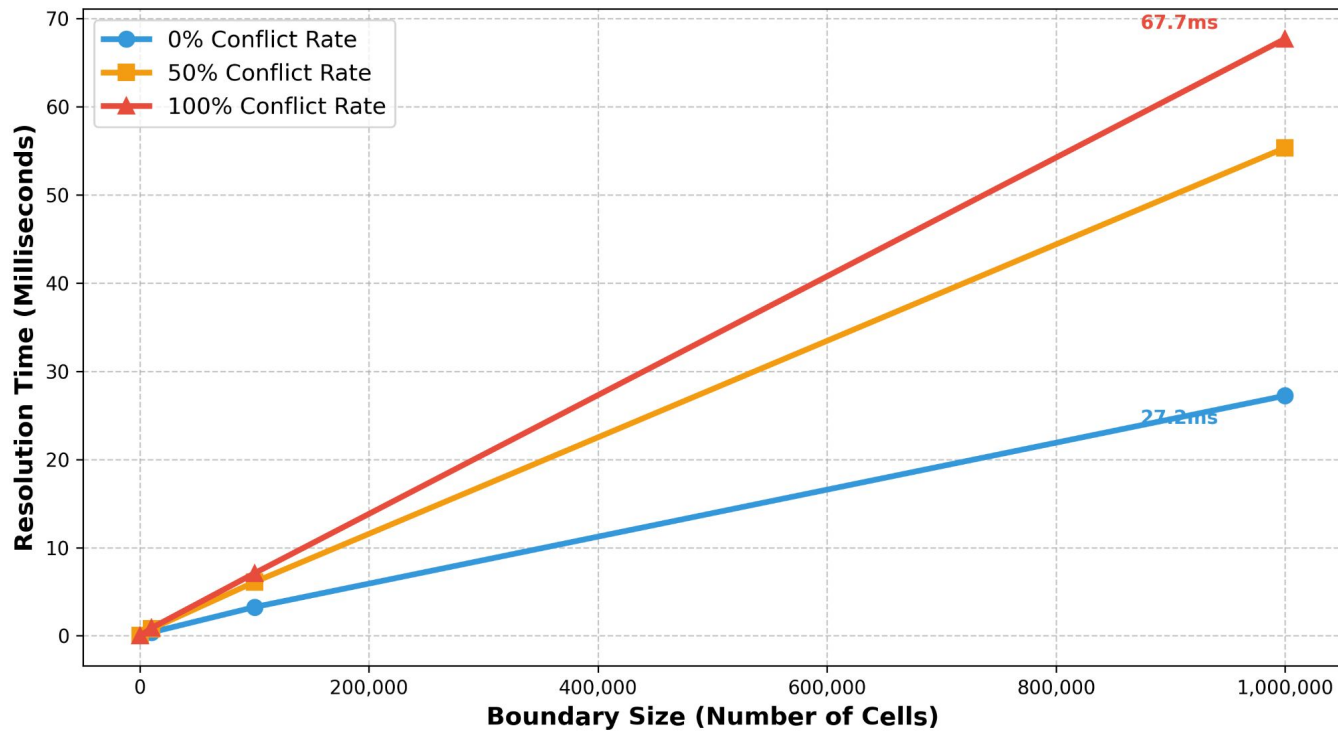
- **Simulation Engines:**
 - **Forest Fire Cellular Automata**
 - **Resolution Protocol:** FIRE > TREE > EMPTY
 - **Limitation:** No temporal causality
 - **Warehouse Fleet Routing**
 - **States:** EMPTY, ROBOT, PACKAGE/WALL
 - Robots optimistically path find to target. At boundary: Freezes physically and speculates pathfinding
 - **Resolution Protocol:** If the network reveals an optimistic path hit a wall, the node instantly flushes the robot's speculative path array and bumps it backward

Results



Results

Algorithm Scalability: True Linear Time $O(N)$



Conclusion

- **Trade-Off:** Speculative computation vs network bottlenecks. Wasting compute on one boundary conflict saves the throughput for the rest of the grid.
- **Asynchronous Compute:** Decoupling network I/O and compute allows nodes to execute logic in parallel with network lag.
- **Generalization:** Generalizes to any high-frequency spatial system where total throughput out-prioritizes strict local causality

Questions?